A-A-51138
11 January 1985

COMMERCIAL ITEM DESCRIPTION

BRAZING AND SOLDERING SET (ACETYLENE GAS)

The General Services Administration has authorized the use of this commercial item description in preference to Federal Specification GGG-B-00690.

This description covers a brazing and soldering set designed to burn acetylene gas and ambient air to produce a flame suitable for light brazing.

Salient characteristics. The brazing and soldering set shall consist of a pressure regulator, torch body, torch tips, soldering iron tip, hose assembly, adapter, wrench, carrying case, and any other item which the manufacturer may include in his standard set.

Pressure regulator. The pressure regulator shall conform to Underwriters Laboratory Standard UL252. The inlet connector shall be designed to fit a type "B" acetylene tank with a fitting conforming to the Compressed Gas Association (CGA) Standard V1-77. The outlet connector shall be a type "A" conforming to Compressed Gas Association Pamphlet E-1. The regulator shall have a pressure gauge with minimal calibration readings of 1/4, 1/2, 3/4, and full. The pressure shall be adjustable by means of a hand-operated device and shall limit the output pressure to a maximum of 15 PSI.

Torch body. One end of the torch body shall be designed to receive the torch tips and the opposite end shall have a standard "A" size hose connection. The torch body shall be furnished with a cut-off valve. There shall be no leakage of gas from the tip and hose connections when they are properly tightened. The material used in the construction of the torch body shall be of the type and quality normally used for the purpose in commercial practice.

Torch tips. The torch tips shall be made of brass or bronze. The tip flame port shall conform to table I and the opposite end shall be compatible with the torch body. One torch tip of each size identified in table I shall be included in the set.

TABLE I. Flame port size.

Tip size	Minimum port size	Maximum port size
No. 1 (fine)	0.075 in. (2.0 mm)	0.112 in. (3.0 mm)
No. 2 (small)	0.130 in. (3.5 mm)	0.170 in. (4.5 mm)
No. 4 (medium)	0.185 in. (4.5 mm)	0.215 in. (5.5 mm)
No. 5 (large)	0.240 in. (6.0 mm)	0.260 in. (6.5 mm)
No. 8 (soldering iron)	0.080 in. (2.0 mm)	0.250 in. (6.5 mm)

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Soldering iron tip. The soldering iron tip shall be a pointed (pyramid) type and manufactured from copper. The soldering iron tip flame port shall conform to table I.

Hose assembly. The hose assembly shall have at least twelve feet of 3/16 inch diameter rubber acetylene gas grade hose with "A" size fittings attached to each end which are compatible with the regulator outlet and torch body input.

Adapter. The adapter shall make connection from a 10 cubic foot "MC" tank to the 40 cubic foot "B" tank pressure regulator. The fittings shall conform to CGA-V1-77.

Wrench. The wrench shall accommodate the type "B" and "MC" tank valves, adapter, pressure regulator and "A" size hose nuts.

Carrying case. The carrying case shall be made of metal or plastic and shall be of sufficient size to hold all components contained in the set.

Flint lighter. The flint lighter shall contain a replaceable flint and extra flints.

Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the commercial item description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Acceptance inspection. The supplier shall randomly select the number of samples specified in table II that is applicable to the quantity of items being purchased. Each sample shall be visusally inspected, measured, and tested, by the supplier, to determine conformance with the salient characteristics specified herein. Any sample failing to meet these characteristics shall be considered a defective item. If the number of defective items equals, or is less than the quantity specified under "Acceptance, number defective" column of table III, the supplier shall replace the defective samples with acceptable items and the entire lot shall be considered acceptable for shipment. If the number of defective samples equals, or exceeds, the quantity specified in the "Rejection, number defective" column of table II, the entire lot shall be considered defective and unsuitable for acceptance.

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TABLE II. Sampling plan.

Lot size	Sample size	Acceptance, number defective	Rejection, number defective
2 to 8,	2	0	1
9 to 15	3	0	1
16 to 25	5	0	1
26 to 50	8	1	2
51 to 90	13	1	2
91 to 150	20	2	3
151 to 280	32	3	4
181 to 500	50	5	6

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with ASTM D 3951. When specified in the contract or order, preservation, packaging, packing, labeling, and marking shall be in accordance with MIL-W-45562, level as specified.

Notes. Purchaser shall specify quantity of sets required.

UL Standard 252 is available from Underwriters Laboratories, Inc., Publication Stock, 333 Pfingsten Road, Northbrook, IL 60062.

American Society for Testing and Materials (ASTM) Standard D 3951 is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.

Compressed Gas Association, Inc., Pamphlet E-1 and Standard V1-77 are available from Compressed Gas Association, Inc., 500 5th Ave., New York, NY 10036.

Copies of military specifications required by manufacturers in connection with specific acquisitions should be obtained from the contracting activity or as directed by the contracting officer.

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MILITARY INTERESTS:

Custodians

Army - AL
Air Force - 99

Review Activities

Army - AL
Air Force - 99
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITIES:

NASA - JFK
GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3433-0101

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